

Clinical Risk Assessment Report: Patient 96

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 57.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.57x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +7.7%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 1.03 [Top 10%] (+7.7%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.36 [Bottom 11%] (+6.2%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.89 [Top 15%] (+5.0%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.98 [Top 14%] (+4.6%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.24 [Bottom 6%] (+3.0%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.17 [Top 12%] (+3.0%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.01 [Top 15%] (+2.9%)**
Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.64 [Top 20%] (+2.7%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

